In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set()
```

In [6]:
```
df = pd.read_csv("Video_Games_Sales_as_at_22_Dec_2016.csv")
df.head(-1)
```

Out[6]:

	Name	Platform	Year_of_Release	Genre	Publisher	NA_Sales	EU_Sale
0	Wii Sports	Wii	2006.0	Sports	Nintendo	41.36	28.9
1	Super Mario Bros.	NES	1985.0	Platform	Nintendo	29.08	3.5
2	Mario Kart Wii	Wii	2008.0	Racing	Nintendo	15.68	12.7
3	Wii Sports Resort	Wii	2009.0	Sports	Nintendo	15.61	10.9
4	Pokemon Red/Pokemon Blue	GB	1996.0	Role-Playing	Nintendo	11.27	8.8
...	...	...	...	...	...	...	
16713	SCORE International Baja 1000: The Official Game	PS2	2008.0	Racing	Activision	0.00	0.0
16714	Samurai Warriors: Sanada Maru	PS3	2016.0	Action	Tecmo Koei	0.00	0.0
16715	LMA Manager 2007	X360	2006.0	Sports	Codemasters	0.00	0.0
16716	Haitaka no Psychedelica	PSV	2016.0	Adventure	Idea Factory	0.00	0.0
16717	Spirits & Spells	GBA	2003.0	Platform	Wanadoo	0.01	0.0

16718 rows × 16 columns

In [4]:
```
df.dtypes
```

Out[4]:
```
Name                 str
Platform             str
Year_of_Release    float64
Genre                str
Publisher            str
NA_Sales           float64
EU_Sales           float64
JP_Sales           float64
Other_Sales        float64
Global_Sales       float64
Critic_Score       float64
Critic_Count       float64
User_Score           str
User_Count         float64
Developer            str
Rating               str
dtype: object
```

In [20]:
```
df['User_Score'] = pd.to_numeric(df['User_Score'], errors='coerce')
df.dtypes
```

Out[20]:
```
Name                 str
Platform             str
Year_of_Release    float64
Genre                str
Publisher            str
NA_Sales           float64
EU_Sales           float64
JP_Sales           float64
Other_Sales        float64
Global_Sales       float64
Critic_Score       float64
Critic_Count       float64
User_Score         float64
User_Count         float64
Developer            str
Rating               str
dtype: object
```

In [21]:
```
print(df.isna().sum())
```

```
Name                  2
Platform              0
Year_of_Release     269
Genre                 2
Publisher            54
NA_Sales              0
EU_Sales              0
JP_Sales              0
Other_Sales           0
Global_Sales          0
Critic_Score       8582
Critic_Count       8582
User_Score         9129
User_Count         9129
Developer          6623
Rating             6769
dtype: int64
```

In [22]:
```
df.duplicated()[df.duplicated()==True]
```

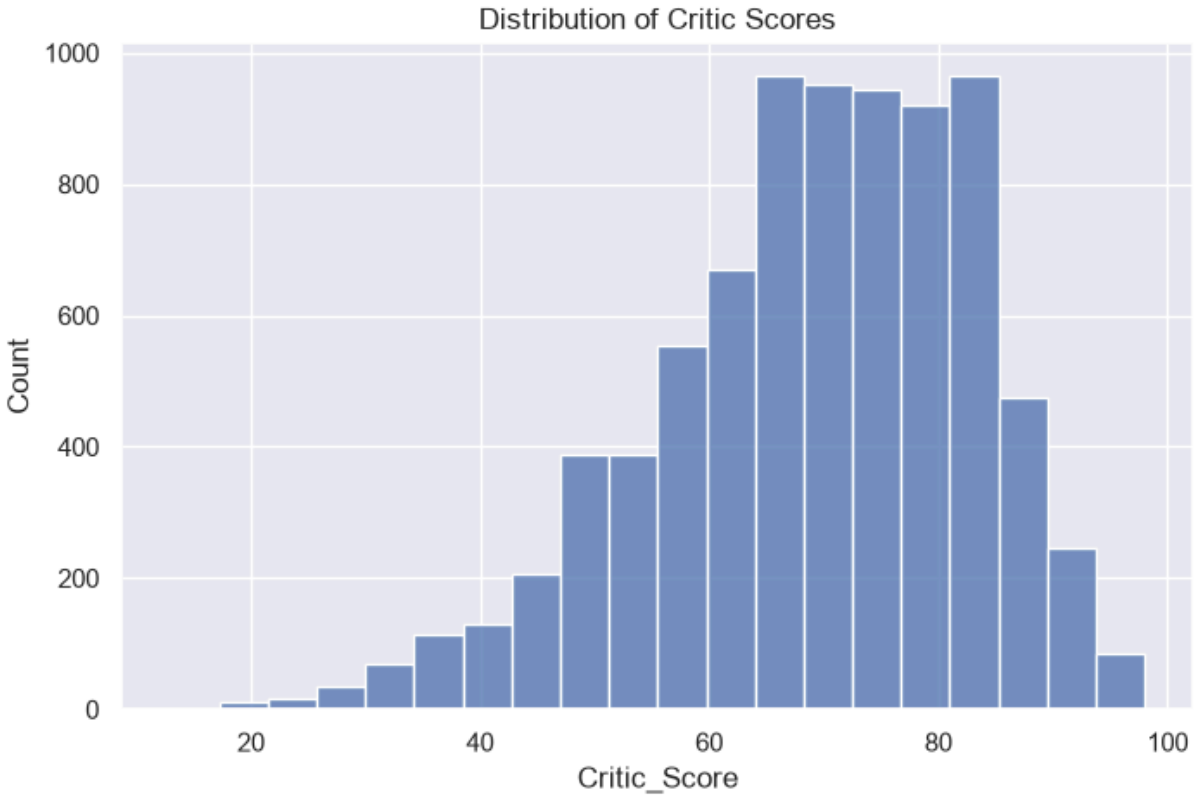
Out[22]:
```
Series([], dtype: bool)
```

In [23]:
```
df.describe(include='all')
```

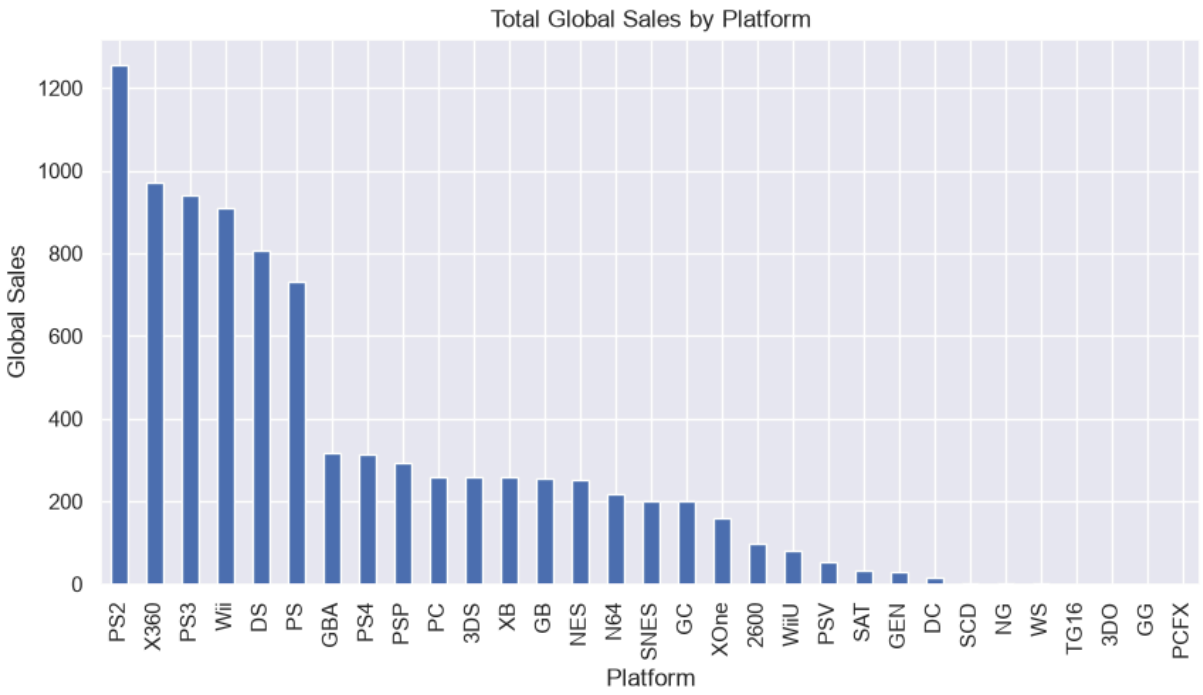
Out[23]:

	Name	Platform	Year_of_Release	Genre	Publisher	NA_Sales	EU_Sales
count	16717	16719	16450.000000	16717	16665	16719.000000	16719.000000
unique	11562	31	NaN	12	581	NaN	NaN
top	Need for Speed: Most Wanted	PS2	NaN	Action	Electronic Arts	NaN	NaN
freq	12	2161	NaN	3370	1356	NaN	NaN
mean	NaN	NaN	2006.487356	NaN	NaN	0.263330	0.145025
std	NaN	NaN	5.878995	NaN	NaN	0.813514	0.503283
min	NaN	NaN	1980.000000	NaN	NaN	0.000000	0.000000
25%	NaN	NaN	2003.000000	NaN	NaN	0.000000	0.000000
50%	NaN	NaN	2007.000000	NaN	NaN	0.080000	0.020000
75%	NaN	NaN	2010.000000	NaN	NaN	0.240000	0.110000
max	NaN	NaN	2020.000000	NaN	NaN	41.360000	28.960000

In [25]:
```
plt.figure(figsize=(8,5))
sns.histplot(df['Critic_Score'], bins=20)
plt.title("Distribution of Critic Scores")
plt.show()
```

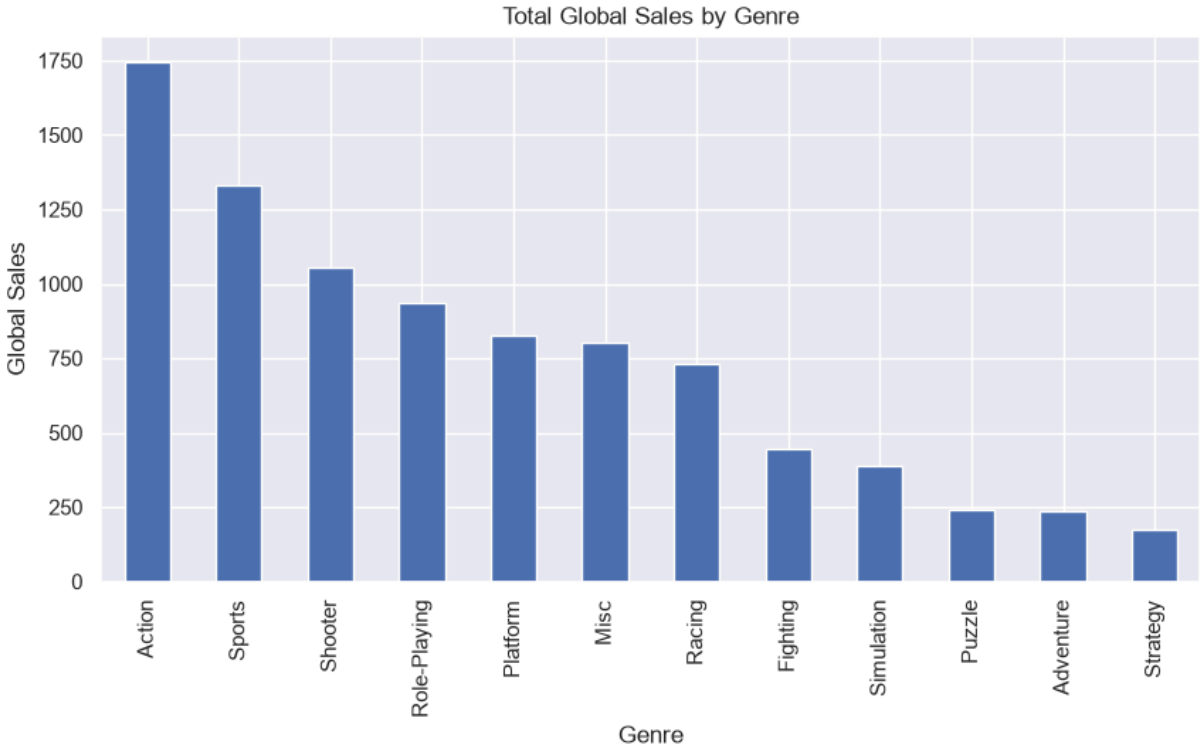


In [27]:
```
platform_sales = df.groupby('Platform')['Global_Sales'].sum().sort_values(ascending
plt.figure(figsize=(10,5))
platform_sales.plot(kind='bar')
plt.title("Total Global Sales by Platform")
plt.ylabel("Global Sales")
plt.show()
platform_sales
```



Out[27]:
```
Platform
PS2     1255.64
X360     971.63
PS3      939.43
Wii      908.13
DS       807.10
PS       730.68
GBA      318.50
PS4      314.23
PSP      294.30
PC       260.30
3DS      259.09
XB       258.26
GB       255.45
NES      251.07
N64      218.88
SNES     200.05
GC       199.36
XOne     159.44
2600      97.08
WiiU      82.16
PSV       54.12
SAT       33.59
GEN       30.78
DC        15.97
SCD        1.87
NG         1.44
WS         1.42
TG16       0.16
3DO        0.10
GG         0.04
PCFX       0.03
Name: Global_Sales, dtype: float64
```

In [28]:
```
genre_sales = df.groupby('Genre')['Global_Sales'].sum().sort_values(ascending=False
plt.figure(figsize=(10,5))
genre_sales.plot(kind='bar')
plt.title("Total Global Sales by Genre")
plt.ylabel("Global Sales")
plt.xlabel("Genre")
plt.show()
genre_sales
```

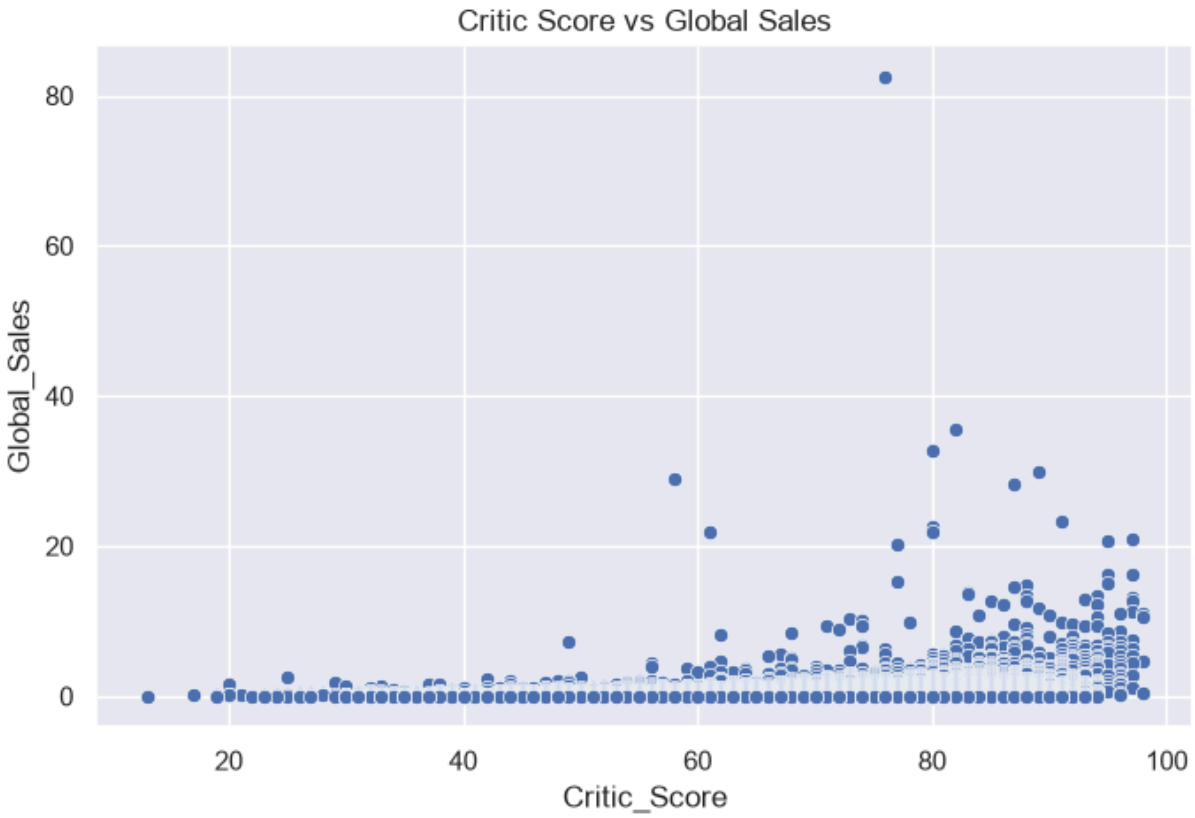


Out[28]:
```
Genre
Action          1745.27
Sports          1332.00
Shooter         1052.94
Role-Playing     934.40
Platform         828.08
Misc             803.18
Racing           728.90
Fighting         447.48
Simulation       390.42
Puzzle           243.02
Adventure        237.69
Strategy         174.50
Name: Global_Sales, dtype: float64
```

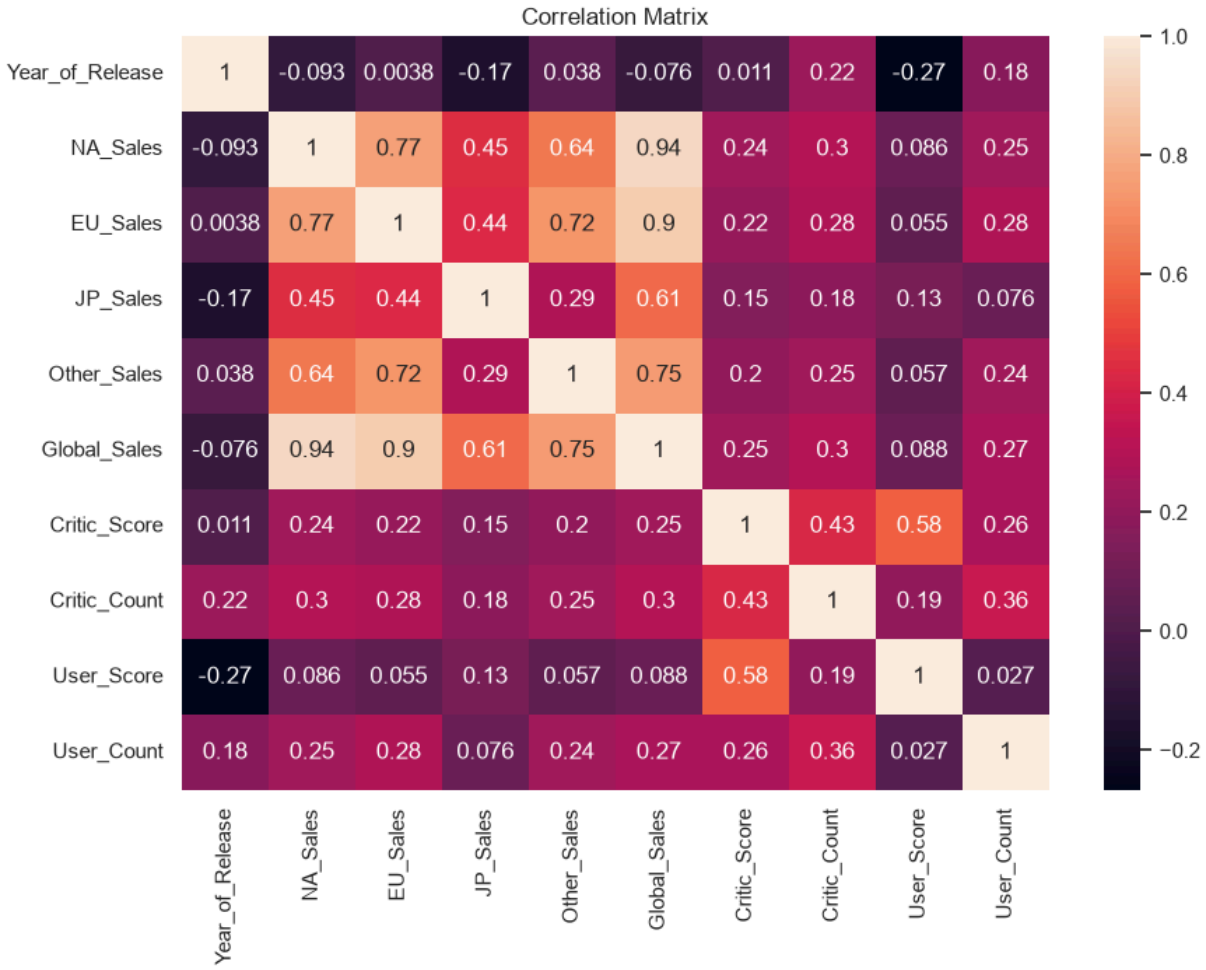
In [30]:
```
publisher_sales = df.groupby('Publisher')['Global_Sales'].sum().sort_values(ascendi
publisher_sales.head(10)
```

Out[30]:
```
Publisher
Nintendo                          1788.81
Electronic Arts                   1116.96
Activision                         731.16
Sony Computer Entertainment        606.48
Ubisoft                            471.61
Take-Two Interactive               403.82
THQ                                338.44
Konami Digital Entertainment       282.39
Sega                               270.35
Namco Bandai Games                 254.62
Name: Global_Sales, dtype: float64
```

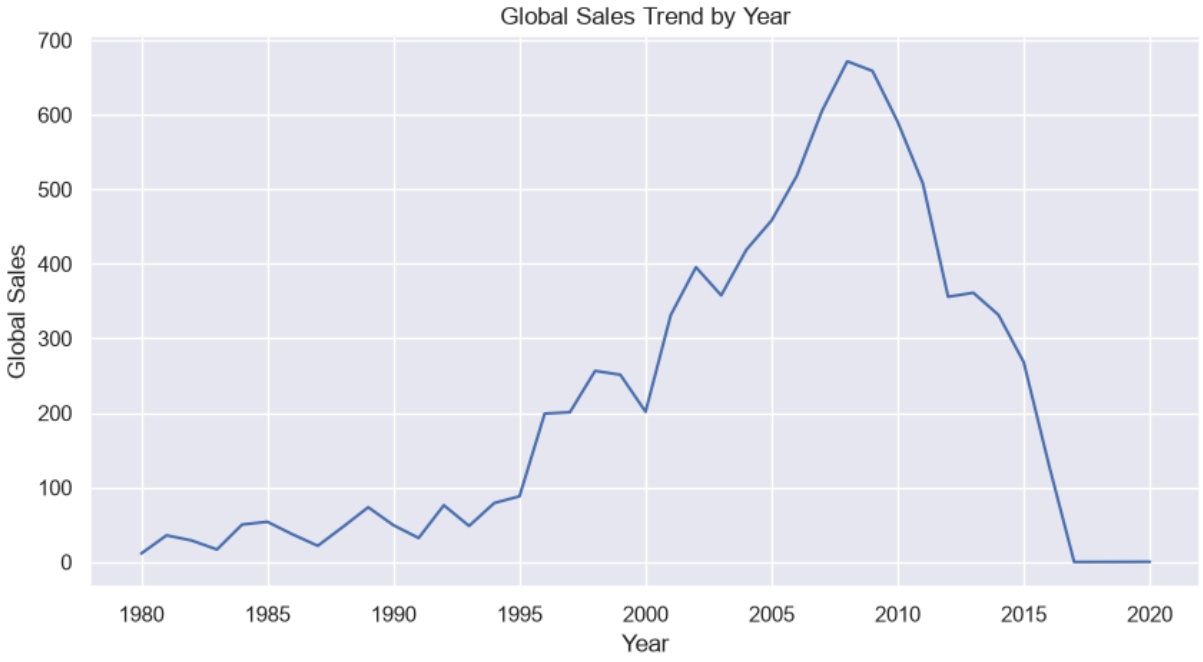
In [31]:
```
plt.figure(figsize=(8,5))
sns.scatterplot(x='Critic_Score', y='Global_Sales', data=df)
plt.title("Critic Score vs Global Sales")
plt.show()
```



In [33]:
```
plt.figure(figsize=(10,7))
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True)
plt.title("Correlation Matrix")
plt.show()
```



In [34]:
```
yearly_sales = df.groupby('Year_of_Release')['Global_Sales'].sum()
plt.figure(figsize=(10,5))
yearly_sales.plot()
plt.title("Global Sales Trend by Year")
plt.ylabel("Global Sales")
plt.xlabel("Year")
plt.show()
```


Global Sales Trend by Year
Global Sales
Year
700
600
500
400
300
200
100
0
1980
1985
1990
1995
2000
2005
2010
2015
2020

In [ ]: